

Prefix Rigidity, Surplus Frontiers, and an Exact Certificate Excluding Collatz 96-Cycles

Nidhal Mghirbi

Independent researcher

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Abstract

Let

$$T(x) = \frac{3x+1}{2^{v_2(3x+1)}}$$

be the accelerated Collatz map on positive odd integers. For a hypothetical positive cycle rotated to its least element x_0 , write S_t for the sum of the first t valuations and put $F_t = S_t - t \log_2 3$. We prove the weighted affine identity

$$x_t = 2^{-F_t} \left(x_0 + \frac{1}{3} \sum_{j < t} 2^{F_j} \right)$$

and the Prefix-Hugging estimate

$$F_t \leq t \log_2 \left(1 + \frac{1}{3x_0} \right),$$

valid also on orbit segments that remain above their initial level. Positive surplus prefixes are therefore confined to one-sided Diophantine near-returns of $\log_2 3$.

Using the published verification floor $x_0 \geq 2^{71}$, we certify every possible positive-surplus time below $q_{28} = 52,449,289,519,716$. The admissible set has 280,139 elements, begins at 72,057,431,991, and has minimum internal spacing 53,715,833. An exact floor-sum oracle computes the same frontier counts in $O(\log N)$ time. We also certify height-independent descent covers for possible cycle minima and obtain the finite height window $2^{71} \leq x_0 < 2^{88}$ for odd-state periods below q_{28} .

As a separate application, Prefix Hugging is inserted into an affine-ladder m -cycle search. Published bounds of Hercher and Simons–de Weger, together with an exact suffix-balanced denominator estimate, first give $K \geq 205,632,218,873,398,596,256$ for a hypothetical 96-cycle. A disjoint 75-branch certificate then excludes every least local minimum in $[2^{71}, 29 \cdot 2^{71}]$. Reapplying the denominator estimate above that window forces $K \geq 7,941,964,418,702,608,664,581$, contradicting the Simons–de Weger upper bound. Hence no nontrivial positive Collatz 96-cycle exists. Every computational decision is made with exact integer or rational arithmetic; the release contains the raw branch logs, source, independent verifier, analytic certificate, and immutable SHA-256 manifest.

Keywords. Collatz problem; accelerated map; cycles; continued fractions; experimental mathematics; computer-assisted proof; exact certificate.

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1 Introduction

The $3x + 1$ problem asks whether every positive integer reaches 1 under the usual Collatz iteration. Its elementary formulation conceals a difficult interaction between multiplication by 3, addition of 1, and powers of 2. Surveys and bibliographies include [6, 7], while cycle restrictions based on continued fractions and linear forms in logarithms were developed, among others, by Steiner, Eliahou, and Simons–de Weger [10, 2, 9].

We work with the accelerated map on odd integers,

$$T(x) = \frac{3x + 1}{2^{v_2(3x+1)}}. \quad (1)$$

A positive accelerated cycle is a sequence of odd integers

$$x_0, x_1, \dots, x_{p-1}, x_p = x_0, \quad 3x_i + 1 = 2^{k_i} x_{i+1}, \quad k_i \geq 1.$$

Rotate a nontrivial cycle so that x_0 is its least element. The resulting minimum circuit has a valuation walk

$$S_t = \sum_{i=0}^{t-1} k_i, \quad F_t = S_t - t\alpha, \quad \alpha = \log_2 3.$$

The full-period product identity says only that F_p is small and positive. The point of this paper is that minimum geometry forces the same smallness at *every prefix*, and on every segment that remains above its starting level.

The endpoint-quotient motivation is as follows. Since a minimum circuit must start with valuation 1, its first maximal run of valuation-one steps can be written

$$k_0 = \dots = k_{m-2} = 1, \quad k_{m-1} = h + 1 \geq 2,$$

and then

$$x_0 = 2^m C - 1$$

for a positive odd endpoint quotient C . Minimality supplies an inequality that is sufficient to exclude the whole branch $2^{m+h} > 3^m$; the endpoint argument is given in section 3.

The main contributions are the following.

- (i) The affine prefix constants admit a weighted form that yields the Prefix Hugging inequality

$$F_t \leq t\lambda_0, \quad \lambda_0 = \log_2 \left(1 + \frac{1}{3x_0} \right).$$

The proof uses only positivity and minimum rotation.

- (ii) Hugging localizes to every clean segment. Consequently, local recurrences above a running level obey the same one-sided Diophantine filter as the full circuit.
- (iii) Combining Hugging with Barina’s verified range through 2^{71} [1], we certify all possible positive-surplus times below q_{28} . This contains the q_{26} range and increases the certified time horizon by more than a factor of eight.
- (iv) The surplus condition is converted into a rotation/floor-sum oracle. It reproduces the exact A_{28} certificate, reconfirms A_{29} by an independent direct scan, and computes frontier counts at much larger convergent heights without enumeration.
- (v) The same framework gives a forced minimum gap, a closed form for the maximum pinning factor, and a certified residue-class descent sieve inside the finite window $x_0 < 2^{88}$.
- (vi) The certified cover data lead to a focused cycle-minimum descent-closure conjecture: endpoint rigidity should leave no indefinitely surviving dyadic residue cylinder for the least element of a nontrivial cycle.

- (vii) Combining the surplus frontier with a suffix-balanced m -cycle reduction and an exact affine-ladder search, we exclude the case $m = 96$. The finite certificate consists of 75 disjoint branches and is checked by an independent raw-log verifier.

Positioning and novelty. This paper combines a certified experimental-mathematics reduction with one complete m -cycle exclusion. Classical continued-fraction and m -cycle methods give strong restrictions on possible cycles [2, 9, 3]; Wang subsequently gave exact affine-ladder exclusions through $m = 95$ [5]. The contributions here are a local affine rigidity principle, a fully reproducible surplus frontier, an $O(\log N)$ structural oracle, residue-class descent certificates, and the discharge of the next affine-ladder case $m = 96$. The broader descent-closure component remains deliberately narrow: it concerns possible cycle minima, not arbitrary trajectories. Every finite statement is paired with an exact certificate and an independently checkable proof contract.

The frontier and descent results are necessary conditions for hypothetical cycles; theorem 11.3 is a complete exclusion of one m -cycle case. The paper does not prove the Collatz conjecture or eliminate all endpoint quotients. In particular, the negative first-spike branch and periods at least q_{28} remain open outside the $m = 96$ argument.

2 Affine coordinates and product defect

Definition 2.1. For a finite valuation word $w = (k_0, \dots, k_{p-1})$, put

$$S_0 = 0, \quad S_t = \sum_{i=0}^{t-1} k_i, \quad A_t = \sum_{j=0}^{t-1} 3^{t-1-j} 2^{S_j}.$$

Thus $A_0 = 0$, and $A_{t+1} = 3A_t + 2^{S_t}$.

Lemma 2.2 (Affine prefix identity). *For every realized orbit prefix,*

$$2^{S_t} x_t = 3^t x_0 + A_t. \tag{2}$$

In particular, a closed word with p odd states satisfies

$$(2^{S_p} - 3^p) x_0 = A_p. \tag{3}$$

Proof. The assertion is clear for $t = 0$. If it holds at t , then

$$2^{S_{t+1}} x_{t+1} = 3 \cdot 2^{S_t} x_t + 2^{S_t} = 3^{t+1} x_0 + 3A_t + 2^{S_t} = 3^{t+1} x_0 + A_{t+1}.$$

Setting $t = p$ and $x_p = x_0$ gives (3). □

Equation (3) displays the familiar self-generated denominator $2^{S_p} - 3^p$. It gives a finite test for a proposed valuation necklace: if the denominator is positive and divides A_p , the resulting integer states must still be checked for positivity, oddness, exact valuations, and primitivity.

Lemma 2.3 (Product defect). *For a positive accelerated cycle,*

$$S_p - p\alpha = \sum_{i=0}^{p-1} \log_2 \left(1 + \frac{1}{3x_i} \right) > 0. \tag{4}$$

If x_0 is the least cycle element, then

$$0 < S_p - p\alpha < \frac{p}{3x_0 \log 2}. \tag{5}$$

Proof. Write

$$2^{k_i} x_{i+1} = 3x_i \left(1 + \frac{1}{3x_i}\right)$$

and multiply around the cycle. Cancellation of the states gives (4). Since $x_i \geq x_0$ and $\log(1+u) < u$ for $u > 0$, summing the natural logarithms gives (5). $\square$

Theorem 2.4 (Rational product window). *Let a positive accelerated cycle have p odd states, total valuation S_p , minimum x_0 , and defect*

$$D = 2^{S_p} - 3^p > 0.$$

Then

$$x_0 < \frac{p2^{S_p}}{3D}. \quad (6)$$

Thus, for fixed p and S_p , the possible cycle minimum lies in an explicit finite rational interval.

Proof. By theorem 2.3,

$$\log\left(\frac{2^{S_p}}{3^p}\right) < \frac{p}{3x_0}.$$

For $u > 0$, $\log(1+u) > u/(1+u)$. Taking $u = D/3^p$ therefore gives

$$\frac{D}{2^{S_p}} < \log\left(1 + \frac{D}{3^p}\right) < \frac{p}{3x_0},$$

which rearranges to (6). $\square$

3 Endpoint quotients and the first-spike gate

Definition 3.1. A *primitive minimum circuit* is a nontrivial positive accelerated cycle rotated so that

$$x_i \geq x_0 > 1 \quad (0 \leq i < p), \quad x_i \neq x_0 \quad (0 < i < p),$$

and the displayed period is primitive.

Lemma 3.2 (Endpoint form). *In a primitive minimum circuit, $k_0 = 1$. If*

$$k_0 = \cdots = k_{m-2} = 1, \quad k_{m-1} = h + 1 \geq 2,$$

then $m \geq 2$ and there is a positive odd integer C such that

$$x_0 = 2^m C - 1, \quad (7)$$

$$x_t = 3^t 2^{m-t} C - 1 \quad (0 \leq t < m), \quad (8)$$

$$x_m = \frac{3^m C - 1}{2^h}. \quad (9)$$

Proof. If $k_0 \geq 2$, then $x_1 \leq (3x_0 + 1)/4 < x_0$, contradicting minimality. Iterating $x \mapsto (3x + 1)/2$ backward through the maximal initial run yields (7)–(8); the last valuation gives (9). Oddness of x_0 forces C odd. $\square$

Corollary 3.3 (Endpoint-quotient product bound). *Under the hypotheses of theorem 3.2, a closed circuit satisfies*

$$C < 2^{-m} \left(1 + \frac{p2^{S_p}}{3(2^{S_p} - 3^p)}\right).$$

Hence fixed integers (p, S_p, m) leave only finitely many endpoint quotients C to test.

Proof. Combine $x_0 = 2^m C - 1$ with theorem 2.4. $\square$

The branch $2^{m+h} > 3^m$ will be called the *positive first-spike branch*, because the coefficient of C is positive after the minimum inequality is rearranged. The following statement corrects the logical form of the gate: minimality gives “ $\leq$ ”, not an equality.

Theorem 3.4 (First-spike gate). *There are no integers $m \geq 2$, $h \geq 1$, and $C \geq 1$ satisfying*

$$2^{m+h} > 3^m, \quad C(2^{m+h} - 3^m) \leq 2^h - 1. \quad (10)$$

Consequently, the first spike of every primitive minimum circuit lies in the branch

$$3^m > 2^{m+h}.$$

Proof. Minimality at time m gives $x_m \geq x_0$. Using (7) and (9) yields precisely the second inequality in (10). Since $C \geq 1$, a solution would satisfy

$$\frac{3^m}{2^m} < 2^h \leq \frac{3^m - 1}{2^m - 1}. \quad (11)$$

Put

$$w_m = \log_2 \frac{1 - 3^{-m}}{1 - 2^{-m}}.$$

Then (11) places the integer h in

$$(m(\alpha - 1), m(\alpha - 1) + w_m].$$

The distance from the left endpoint to the next integer is

$$\delta^+(m) := \lceil m\alpha \rceil - m\alpha.$$

Thus a solution requires $\delta^+(m) \leq w_m$.

Rhin’s universal two-logarithm estimate [8] implies that, for $m \geq 2$,

$$\|m\alpha\|_{\mathbb{Z}} \geq \frac{2^{-13.3}}{\log 2} m^{-13.3}. \quad (12)$$

Indeed, apply the estimate to $r \log 2 - m \log 3$, where r is a nearest integer to $m\alpha$ and $\max\{r, m\} \leq 2m$. On the other hand,

$$w_m < \frac{2^{-m}}{(1 - 2^{-m}) \log 2}. \quad (13)$$

For $m \geq 103$, the right side of (13) is smaller than the right side of (12). Two fully exact checks are

$$2^{897} > 2 \cdot 103^{133}, \quad 2(2^{103} - 1)^{10} > 2^{1030}.$$

The second inequality is exactly $(1 - 2^{-103})^{10} > 1/2$; together, after raising the comparison to the tenth power, the two inequalities verify the claim at $m = 103$. The logarithm of the ratio

$$R(m) = \frac{2^{13.3-m} m^{13.3}}{1 - 2^{-m}}$$

has derivative

$$\frac{R'(m)}{R(m)} = \frac{13.3}{m} - \log 2 - \frac{(\log 2)2^{-m}}{1 - 2^{-m}} < 0 \quad (m \geq 103).$$

Hence $w_m < \|m\alpha\|_{\mathbb{Z}} \leq \delta^+(m)$ for all $m \geq 103$.

The finite range $2 \leq m \leq 102$ is checked by exact integer arithmetic in the supplement: for each m , every h satisfying the upper inequality in (11) is tested against the strict lower inequality. No case survives. Therefore (10) has no solution. $\square$

Corollary 3.5 (The $m = 2$ endpoint residue). *Suppose a primitive minimum circuit has $m = 2$. Then*

$$h = 1, \quad C \equiv 7 \pmod{8}, \quad x_0 \equiv 27 \pmod{32}.$$

Proof. Here $h = v_2(9C - 1)$. If $C \equiv 1 \pmod{4}$, then $h \geq 2$ and $2^{2+h} > 9$, contrary to theorem 3.4. Hence $C \equiv 3 \pmod{4}$ and $h = 1$. Now $x_2 = (9C - 1)/2$. If the next valuation were at least 2, then

$$x_3 \leq \frac{27C - 1}{8} < 4C - 1 = x_0,$$

contradicting minimality. Thus $v_2(27C - 1) = 2$, which is equivalent to $C \equiv 7 \pmod{8}$; substituting in $x_0 = 4C - 1$ gives the final congruence. $\square$

Corollary 3.6 (Further branching in the $m = 2$ stratum). *Under the hypotheses of theorem 3.5, put*

$$x_3 = \frac{27C - 1}{4}, \quad k_3 = v_2(3x_3 + 1).$$

Exactly one of the following two residue alternatives is possible:

$$C \equiv 7 \pmod{16}, \quad k_3 = 1,$$

or

$$C \equiv 31 \pmod{32}, \quad k_3 = 2.$$

In particular, the class $C \equiv 15 \pmod{32}$ is impossible.

Proof. From theorem 3.5, the valuation following $x_2 = (9C - 1)/2$ is one, so $x_3 = (27C - 1)/4$. Since

$$3x_3 + 1 = \frac{81C + 1}{4},$$

a value $k_3 \geq 3$ would imply

$$x_4 \leq \frac{81C + 1}{32} < 4C - 1 = x_0,$$

contrary to minimality. Thus $k_3 \leq 2$. If $C \equiv 7 \pmod{16}$, then $81C + 1 \equiv 8 \pmod{16}$, hence $k_3 = 1$. In the remaining class $C \equiv 15 \pmod{16}$, one has $v_2(81C + 1) \geq 4$; the constraint $k_3 \leq 2$ forces $v_2(81C + 1) = 4$. Modulo 32 this is $17C \equiv 15$, hence $C \equiv 31 \pmod{32}$. The other lift $C \equiv 15 \pmod{32}$ would give $k_3 \geq 3$ and is excluded. $\square$

4 Weighted affine mass and Prefix Hugging

For the rest of the paper, set

$$\alpha = \log_2 3, \quad F_t = S_t - t\alpha, \quad \lambda_0 = \log_2 \left(1 + \frac{1}{3x_0} \right),$$

and define the weighted prefix mass

$$\sigma_t = \sum_{j=0}^{t-1} 2^{F_j}, \quad \sigma_0 = 0.$$

Lemma 4.1 (Weighted affine mass). *For $t \geq 1$,*

$$A_t = 3^{t-1} \sigma_t,$$

and for every $t \geq 0$,

$$x_t = 2^{-F_t} \left(x_0 + \frac{\sigma_t}{3} \right). \tag{14}$$

Proof. Since $2^{S_j} = 3^j 2^{F_j}$,

$$A_t = \sum_{j < t} 3^{t-1-j} 2^{S_j} = 3^{t-1} \sum_{j < t} 2^{F_j}.$$

Substitution into (2), followed by division by $2^{S_t} = 3^t 2^{F_t}$, gives (14). $\square$

Theorem 4.2 (Prefix Hugging). *For every prefix $0 \leq t \leq p$ of a primitive minimum circuit,*

$$F_t \leq t\lambda_0 < \frac{t}{3x_0 \log 2}, \quad (15)$$

$$\sigma_t \leq 3x_0 \left[\left(1 + \frac{1}{3x_0} \right)^t - 1 \right]. \quad (16)$$

Proof. Minimum rotation gives $x_t \geq x_0$. By (14),

$$2^{F_t} \leq 1 + \frac{\sigma_t}{3x_0}.$$

Consequently

$$\sigma_{t+1} = \sigma_t + 2^{F_t} \leq 1 + \left(1 + \frac{1}{3x_0} \right) \sigma_t.$$

Solving this affine recursion from $\sigma_0 = 0$ gives (16). Substituting back yields $2^{F_t} \leq (1 + 1/(3x_0))^t$, which is (15); the strict upper bound follows from $\log(1 + u) < u$. $\square$

Corollary 4.3 (Zero overshoot). *If $F_t > 0$ and $t\lambda_0 < 1$, then*

$$S_t = \lceil t\alpha \rceil.$$

Proof. The positive integer displacement above $t\alpha$ is less than 1 by theorem 4.2, so it is the least possible displacement. $\square$

Corollary 4.4 (Height–walk dictionary). *For every $0 \leq t \leq p$,*

$$x_0 2^{-F_t} \leq x_t \leq x_0 2^{-F_t + t\lambda_0}. \quad (17)$$

Equivalently,

$$-\log_2 \frac{x_t}{x_0} \leq F_t \leq -\log_2 \frac{x_t}{x_0} + t\lambda_0.$$

Proof. The lower bound follows from $\sigma_t \geq 0$ in (14); the upper bound follows from (16). $\square$

5 Localization to clean orbit segments

The Hugging argument is not tied to the global minimum.

Definition 5.1. Fix a time a . A segment of length s is *clean above x_a* if

$$x_{a+j} \geq x_a \quad (0 \leq j \leq s).$$

For such a segment put

$$S_j^{(a)} = S_{a+j} - S_a, \quad F_j^{(a)} = S_j^{(a)} - j\alpha, \quad \lambda_a = \log_2 \left(1 + \frac{1}{3x_a} \right),$$

and

$$\sigma_j^{(a)} = \sum_{r=0}^{j-1} 2^{F_r^{(a)}}.$$

Theorem 5.2 (Segment Hugging). *If a segment of length s is clean above x_a , then for $0 \leq j \leq s$,*

$$F_j^{(a)} \leq j\lambda_a, \quad (18)$$

$$\sigma_j^{(a)} \leq 3x_a \left[\left(1 + \frac{1}{3x_a} \right)^j - 1 \right]. \quad (19)$$

If the segment stays above x_a only for $0 \leq j < s$ and lands below at $j = s$, the same statements hold for all strict pre-landing prefixes.

Proof. The shifted affine identity is

$$x_{a+j} = 2^{-F_j^{(a)}} \left(x_a + \frac{\sigma_j^{(a)}}{3} \right).$$

For each clean prefix, $x_{a+j} \geq x_a$, and the proof of theorem 4.2 repeats verbatim with x_a in place of x_0 . $\square$

This localization is useful because any member of a hypothetical cycle is at least its global minimum. Hence every clean segment inherits the same external verification floor used below.

6 Near-return geometry

Definition 6.1. A time t is a *positive-surplus time* if $F_t > 0$. Write

$$\mathcal{S} = \{1 \leq t \leq p : F_t > 0\}, \quad \delta^+(t) = \lceil t\alpha \rceil - t\alpha.$$

Lemma 6.2 (Rotation form of the surplus). *Let $\theta = 2 - \alpha = \log_2(4/3)$. Then the surplus is a one-sided rotation orbit,*

$$\delta^+(t) = \{t\theta\} \quad (t \geq 1),$$

where $\{\cdot\}$ denotes the fractional part; in particular $\|t\theta\|_{\mathbb{Z}} = \|t\alpha\|_{\mathbb{Z}}$. Thus a positive-surplus candidate is exactly an integer t for which $t\theta$ lies just above an integer, and the admissible times are governed by the three-distance theorem and the Ostrowski expansion of θ .

Proof. Because $t\alpha \notin \mathbb{Z}$ for $t \geq 1$, we have $\lceil t\alpha \rceil - t\alpha = \{-t\alpha\}$, and $-\alpha \equiv \theta \pmod{1}$ gives $\{-t\alpha\} = \{t\theta\}$. Finally $t\theta = 2t - t\alpha$ and $\|-x\|_{\mathbb{Z}} = \|x\|_{\mathbb{Z}}$ yield $\|t\theta\|_{\mathbb{Z}} = \|t\alpha\|_{\mathbb{Z}}$. $\square$

Theorem 6.3 (Bohr rigidity). *For every $t \in \mathcal{S}$,*

$$\delta^+(t) \leq F_t \leq t\lambda_0. \quad (20)$$

If $t < t'$ lie in $\mathcal{S}$ and $g = t' - t$, then

$$\|g\alpha\|_{\mathbb{Z}} < t'\lambda_0. \quad (21)$$

Consequently, the effective estimate (12) gives

$$g > \left(\frac{2^{-13.3}}{t'\lambda_0 \log 2} \right)^{1/13.3}.$$

Proof. The left side of (20) is the least possible positive difference between an integer and $t\alpha$; the right side is theorem 4.2. Also

$$F_{t'} - F_t = (S_{t'} - S_t) - g\alpha.$$

Since both F_t and $F_{t'}$ are positive, $|F_{t'} - F_t| < \max(F_t, F_{t'}) \leq t'\lambda_0$, proving (21). The last assertion follows from (12). $\square$

Theorem 6.4 (Continued-fraction localization). *Let P_j/Q_j be the convergents of α . If $t \in S$ and*

$$t < (2\lambda_0)^{-1/2},$$

then the reduced fraction associated with S_t/t is an upper convergent of α . Thus for some j and some integer $r \geq 1$,

$$(t, S_t) = r(Q_j, P_j).$$

Moreover,

$$Q_j(Q_j + Q_{j+1}) > \lambda_0^{-1}.$$

Proof. By (20),

$$0 < \frac{S_t}{t} - \alpha \leq \lambda_0 < \frac{1}{2t^2}.$$

Legendre's criterion implies that the reduced fraction is a convergent, and the sign makes it an upper convergent. Finally, $|Q_j\alpha - P_j| \leq Q_j\lambda_0$, whereas the standard lower bound for a convergent is $|Q_j\alpha - P_j| > 1/(Q_j + Q_{j+1})$. $\square$

7 The exact certified frontier below q_{28}

Barina has verified convergence of every positive integer below 2^{71} [1]. Therefore every member of a nontrivial positive cycle satisfies

$$x_i \geq x_0 \geq X := 2^{71}. \quad (22)$$

Set

$$M = 3X \log 2. \quad (23)$$

We index continued fractions from 0. Exact rational interval arithmetic certifies

$$\alpha = [1; 1, 1, 2, 2, 3, 1, 5, 2, 23, 2, 2, 1, 1, 55, 1, 4, 3, 1, 1, 15, 1, 9, 2, 5, 7, 1, 1, 4, \dots].$$

The relevant denominator data are

j	P_j	Q_j
21	10,439,860,591	6,586,818,670
22	103,768,467,013	65,470,613,321
23	217,976,794,617	137,528,045,312
24	1,193,652,440,098	753,110,839,881
25	8,573,543,875,303	5,409,303,924,479
26	9,767,196,315,401	6,162,414,764,360
27	18,340,740,190,704	11,571,718,688,839
28	83,130,157,078,217	52,449,289,519,716

For $j = 28$,

$$\varepsilon = Q_{28}\alpha - P_{28} > 0.$$

Definition 7.1 (Certified admissible set). Define

$$\mathcal{A}_{28} = \left\{ 1 \leq t < Q_{28} : \delta^+(t) < \frac{t}{M} \right\}. \quad (24)$$

This is a necessary filter for positive-surplus times, not a sufficient condition for a cycle.

Theorem 7.2 (Universal surplus filter). *Let $t < Q_{28}$ be a positive-surplus time in a primitive minimum circuit. Then*

$$t \in \mathcal{A}_{28}, \quad S_t = \lceil t\alpha \rceil.$$

The same conclusion holds for a positive-surplus endpoint of any clean segment of length less than Q_{28} .

Proof. By theorem 4.2 and (22),

$$0 < \delta^+(t) \leq F_t < \frac{t}{3x_0 \log 2} \leq \frac{t}{M}.$$

Thus $t \in \mathcal{A}_{28}$. The exact rational certificate also verifies $Q_{28}/M < 1$, so $F_t < t/M < 1$ and theorem 4.3 applies. For a clean segment, use theorem 5.2 and $x_a \geq x_0 \geq X$. $\square$

Theorem 7.3 (Exact A_{28} certificate). *The set (24) contains exactly 280,139 integers. Its least and greatest elements are*

$$\begin{aligned} \min \mathcal{A}_{28} &= 72,057,431,991 = Q_{21} + Q_{22}, \\ \max \mathcal{A}_{28} &= 52,449,235,803,883. \end{aligned}$$

Its distribution by convergent bands is

<i>band</i>	<i>number of elements</i>
$[Q_{22}, Q_{23})$	1
$[Q_{23}, Q_{24})$	58
$[Q_{24}, Q_{25})$	2,918
$[Q_{25}, Q_{26})$	890
$[Q_{26}, Q_{27})$	9,770
$[Q_{27}, Q_{28})$	266,502

Moreover, the minimum difference between two distinct elements of $\mathcal{A}_{28}$ is

$$53,715,833 = Q_{16}. \tag{25}$$

The restriction to $t < Q_{26}$ contains exactly 3,867 elements and has maximum 6,159,008,532,722, containing the A_{26} frontier as an exact subcertificate.

Proof. We describe the complete modular certificate. Write $P = P_{28}$, $Q = Q_{28}$, and, for $1 \leq t < Q$,

$$r_t \equiv tP \pmod{Q}, \quad 1 \leq r_t < Q, \quad d_t = Q - r_t.$$

The same exact continued-fraction enclosure gives $0 < Q\varepsilon < 1$. Hence $r_t + t\varepsilon < Q$ for every $t < Q$, and, since $\alpha = P/Q + \varepsilon/Q$, membership in $\mathcal{A}_{28}$ is equivalent to

$$0 < d_t - t\varepsilon < \frac{Qt}{M}. \tag{26}$$

The middle expression is therefore exactly $Q\delta^+(t)$. Every solution of (26) satisfies

$$d_t < t\varepsilon + \frac{Qt}{M} < Q\varepsilon + \frac{Q^2}{M} < 560,278.$$

Thus completeness requires only the scan

$$1 \leq d \leq 560,277.$$

For each d , the congruence

$$tP \equiv -d \pmod{Q}$$

has a unique t in $[1, Q)$ because $\gcd(P, Q) = 1$.

The supplementary verifier encloses $\log 2$ and $\log 3$ by exact rational intervals, hence also encloses ε and $1/M$. It determines both signs in (26) without floating point. Across the entire scan, the exact distance from each of the two decision boundaries is strictly greater than $27/50$. The accepted residues are written in increasing order to `A28_certificate.csv`; a second pass verifies uniqueness, completeness, the band counts, the extrema, and the minimum adjacent gap. $\square$

Theorem 7.4 (Gap quantization). *Every difference between two consecutive elements of $\mathcal{A}_{28}$ is the denominator of a convergent or intermediate fraction (semiconvergent) of α . In particular the minimum spacing (25) is forced: no two positive-surplus times below Q_{28} differ by less than $Q_{16} = 53,715,833$, and this value is attained.*

Proof. Let $t < t'$ be consecutive in $\mathcal{A}_{28}$ and put $g = t' - t$. By (26) both $\delta^+(t)$ and $\delta^+(t')$ lie in $(0, t'/M)$, so theorem 6.2 gives

$$\|g\alpha\|_{\mathbb{Z}} = \|g\theta\|_{\mathbb{Z}} = |\{t'\theta\} - \{t\theta\}| < \frac{t'}{M} < \frac{Q_{28}}{M}.$$

A difference whose orbit distance is this small is, by the theory of best approximation, the denominator of a convergent or semiconvergent of α . For the lower bound, the same exact enclosure that certifies ε and $1/M$ gives

$$\|Q_{15}\alpha\|_{\mathbb{Z}} = |Q_{15}\alpha - P_{15}| > \frac{Q_{28}}{M},$$

while best approximation forces $\|g\alpha\|_{\mathbb{Z}} \geq \|Q_{15}\alpha\|_{\mathbb{Z}}$ for every $1 \leq g < Q_{16}$. Hence no admissible gap is smaller than Q_{16} , and Q_{16} is attained by theorem 7.3. $\square$

Corollary 7.5 (Certified surplus and period gaps). *No positive-surplus time occurs below*

$$72,057,431,991.$$

Any two distinct positive-surplus times below Q_{28} differ by at least 53,715,833. If a primitive minimum circuit has odd-term period $p < Q_{28}$, then

$$p \in \mathcal{A}_{28}, \quad S_p = \lceil p\alpha \rceil, \quad 72,057,431,991 \leq p < Q_{28}.$$

All three conclusions also apply to positive-surplus endpoints within a clean segment, with time measured from the start of that segment.

Proof. The first two assertions follow from theorems 7.2 and 7.3. The full period is a positive-surplus time by theorem 2.3. The segment statement follows from the localized part of theorem 7.2. $\square$

Theorem 7.6 (Continued-fraction lower bound on the cycle length). *Every nontrivial positive cycle of T with least element $x_0 \geq 2^{71}$ has odd-term period*

$$p \geq 72,057,431,991.$$

More generally, for a verification floor $x_0 \geq 2^E$ the period satisfies $p \geq \tau(2^E)$, where

$$\tau(X) = \min \left\{ t \geq 1 : \delta^+(t) < \frac{t}{3X \log 2} \right\}$$

is determined by the continued fraction of α in $O(\log X)$ time:

$$\tau(2^{71}) = 72,057,431,991, \quad \tau(2^{80}) = 2,396,860,564,955, \quad \tau(2^{88}) = 64,021,008,208,555.$$

Proof. The full period p is a positive-surplus time by theorem 2.3, so $\delta^+(p) < p/M$ with $M = 3 \cdot 2^E \log 2$; that is, p is admissible at floor 2^E , and by definition $\tau(2^E)$ is the least admissible time, whence $p \geq \tau(2^E)$. For $E = 71$, theorem 7.3 gives $\min \mathcal{A}_{28} = 72,057,431,991$; since $\mathcal{A}_{28}$ contains *every* admissible time below Q_{28} and any $t \geq Q_{28}$ already exceeds this value, no admissible time is smaller, so $\tau(2^{71}) = 72,057,431,991$. The remaining values are produced and certified by the oracle of section 10 from the convergent and semiconvergent denominators of α . $\square$

Remark 7.7 (Relation to known bounds). Theorem 7.6 is the elementary continued-fraction cycle bound of Crandall and Eliahou [2], recovered here as a direct consequence of the surplus condition; the only new element is the $O(\log X)$ evaluation of τ . It is *not* a record. The m -cycle method gives a stronger unconditional bound: Hercher [3] proves a nontrivial cycle has at least 92 local minima and, with the verification floor (22), at least 1.375×10^{11} odd terms, roughly twice the bound above, with subsequent work pushing the cycle length further into the 10^{12} range. We include τ only because it falls out of the same necessary condition and is computed at no extra cost.

Theorem 7.8 (A finite minimum window below q_{28}). *For $t \in \mathcal{A}_{28}$, define*

$$\rho(t) = \frac{t}{M\delta^+(t)}.$$

The exact certificate proves

$$\rho(t) < 126,669 < 2^{17} \quad (t \in \mathcal{A}_{28}). \quad (27)$$

The maximum is attained precisely at

$$t = jQ_{27}, \quad 1 \leq j \leq 4,$$

and has the closed form

$$\rho_{\max} = \frac{Q_{27}}{M \|Q_{27}\alpha\|_{\mathbb{Z}}} = 126,668.58 \dots$$

Consequently, if a primitive minimum circuit has period $p < Q_{28}$, then

$$2^{71} \leq x_0 < 2^{88}. \quad (28)$$

Equivalently $x_0 < X\rho_{\max} = Q_{27}/(3 \|Q_{27}\alpha\|_{\mathbb{Z}} \log 2) < 2^{88}$, a ceiling fixed by the frontier convergent Q_{27} and independent of the floor X . Thus a future independent verification of convergence for all starting values below 2^{88} would eliminate every nontrivial positive cycle with odd-term period below Q_{28} .

Proof. For a period $p < Q_{28}$, theorem 7.2 gives $S_p = \lceil p\alpha \rceil$, and theorem 2.3 gives

$$\delta^+(p) \log 2 < \frac{p}{3x_0}.$$

Therefore

$$x_0 < \frac{p}{3\delta^+(p) \log 2} = X\rho(p).$$

The exact scan verifies (27), including the stated maximizing times; the closed form for $\rho_{\max}$ follows from $\delta^+(Q_{27}) = \|Q_{27}\alpha\|_{\mathbb{Z}}$ at the maximizer. Since $X = 2^{71}$, (28) follows. The last sentence is conditional on a verification that has not been performed in this work. $\square$

8 Height-independent descent covers

The surplus frontier restricts *when* a minimum circuit can close. A complementary finite certificate can restrict *where* its minimum can lie modulo a power of two. This section records such a residue sieve. It is unconditional: a certified cell descends below its starting value for every integer in the cell, independently of the size of that integer.

Definition 8.1 (Descent cell). Let r be an odd residue modulo 2^n . A tuple (r, n, s, S, c) is a height-independent descent cell if the first s accelerated steps are fixed throughout the cylinder

$$x = r + 2^n u \quad (u \geq 0),$$

the total valuation along this prefix is S , and

$$T^s(r + 2^n u) = c + 3^s 2^{n-S} u \quad (29)$$

for all $u \geq 0$, with

$$c < r, \quad 3^s \leq 2^S.$$

Lemma 8.2 (Cell descent). *Every integer in a height-independent descent cell fails to be a cycle minimum. More precisely, if (r, n, s, S, c) is such a cell, then*

$$T^s(x) < x \quad (x \equiv r \pmod{2^n}, x \geq r).$$

Proof. Write $x = r + 2^n u$. By (29),

$$T^s(x) - x = (c - r) + 2^{n-S}(3^s - 2^S)u.$$

The first term is negative and the second is nonpositive. Thus $T^s(x) < x$. A primitive minimum cannot have a later iterate below itself. $\square$

Theorem 8.3 (Certified primitive-minimum descent cover). *A primitive minimum must satisfy $x_0 \equiv 3 \pmod{4}$. Within this necessary class, the release certifies the following prefix-free depth-24 binary covers. A covered leaf is a descent cell in the sense of theorem 8.1; a residual leaf is simply not claimed to descend by this certificate.*

base class	descent cells	residual classes at depth 24	covered fraction
3 (mod 16)	20	1	1,048,575/1,048,576
7 (mod 16)	12,889	66,815	981,761/1,048,576
11 (mod 16)	12,889	66,815	981,761/1,048,576
15 (mod 16)	39,397	234,067	814,509/1,048,576

The combined covered fraction of the full primitive-minimum class $x_0 \equiv 3 \pmod{4}$ is

$$\frac{1,913,303}{2,097,152} > 0.9123339. \quad (30)$$

Equivalently, only 367,698 residue classes modulo 2^{24} remain from the primitive-minimum class. Consequently, inside the finite window $x_0 < 2^{88}$ of theorem 7.8, the number of candidate minima not already eliminated by these height-independent covers is at most

$$367,698 \cdot 2^{64} = 183,849 \cdot 2^{65} < 2^{83}. \quad (31)$$

Every descent cell used here has at most 14 accelerated steps and total valuation at most 23.

Proof. The congruence $x_0 \equiv 3 \pmod{4}$ follows from $k_0 = 1$ in theorem 3.2. The verifier then checks four residue trees, one for each lift modulo 16. For every listed descent leaf it recomputes the accelerated prefix from the residue alone, verifies that each valuation is forced before the available low bits are exhausted, recomputes the affine data (29), and checks $c < r$ and $3^s \leq 2^S$. By theorem 8.2, every such leaf is excluded.

The verifier also checks that, in each base class, the union of descent and residual leaves is an exact prefix-free tiling of that base class. Therefore the displayed fractions are exact. Summing the four class measures gives (30); refining residual leaves to depth 24 gives 367,698 unresolved classes modulo 2^{24} . Since 2^{24} divides 2^{88} , the count (31) is exact for the interval $[0, 2^{88})$. $\square$

Corollary 8.4 (Endpoint $m = 2$ descent cover). *In the endpoint stratum $m = 2$, theorem 3.5 gives $x_0 \equiv 27 \pmod{32}$. The release certifies a depth-28 prefix-free cover of this class with 173,892 descent cells and 716,705 residual classes. Its covered fraction within $27 \pmod{32}$ is*

$$\frac{7,671,903}{8,388,608} > 0.9145621.$$

Thus, below 2^{88} , the unresolved $m = 2$ endpoint-minimum candidates number at most

$$716,705 \cdot 2^{60} < 2^{80}.$$

Proof. This is the same certificate check as in theorem 8.3, applied to the single base class $27 \pmod{32}$ and depth 28. $\square$

9 A cycle-minimum descent-closure conjecture

The descent certificates above suggest a sharper experimental target than a larger enumeration of admissible periods. A cycle minimum is special: if x_0 is the least element of a nontrivial accelerated cycle, then no later accelerated iterate may fall below x_0 . Thus every height-independent descent cell is an unconditional obstruction to being a cycle minimum. The experiments in this paper indicate that the endpoint and frontier constraints may force all remaining minimum cylinders into such descent cells after sufficiently fine 2-adic refinement.

For clarity, the present certified evidence is summarized in the form used by the conjecture. The first row is the global primitive-minimum sieve after the necessary congruence $x_0 \equiv 3 \pmod{4}$; the second row is the endpoint $m = 2$ stratum after the stronger congruence $x_0 \equiv 27 \pmod{32}$.

modulus/depth	class family	covered fraction	residual classes
2^{24}	primitive minima, $x_0 \equiv 3 \pmod{4}$	1,913,303/2,097,152	367,698
2^{28}	endpoint $m = 2$, $x_0 \equiv 27 \pmod{32}$	7,671,903/8,388,608	716,705

Let $\mathcal{D}_R$ denote the union of dyadic residue classes modulo 2^R that are refined by certified height-independent descent cells. For a continued-fraction frontier q_k , let $\mathcal{B}_k(R)$ denote the dyadic classes modulo 2^R that remain compatible with being the least element of a nontrivial accelerated cycle of odd period $p < q_k$ after imposing the exact minimum, endpoint-quotient, and prefix-frontier constraints of the present method, but before declaring local descent cells excluded.

Conjecture 9.1 (Cycle-minimum descent closure). For every continued-fraction frontier q_k there is a dyadic depth $R(k)$ such that

$$\mathcal{B}_k(R(k)) \subseteq \mathcal{D}_{R(k)}.$$

Equivalently, no residue cylinder compatible with the endpoint/frontier necessary conditions survives indefinitely as a possible least element of a nontrivial accelerated Collatz cycle.

A proof of Conjecture 9.1 at a single frontier q_k would rule out nontrivial cycles with odd period below q_k . A proof for all frontiers would establish the cycle-exclusion part of the Collatz conjecture. The point of the conjecture is not that the present depth-24 and depth-28 covers already close the problem; they do not. Rather, the data identify a concrete mechanism: endpoint rigidity appears to compress possible minima into residue families that are progressively exhausted by finite local descent certificates. This makes the next experimental problem precise: refine the residual classes, preferably using the logarithmic frontier oracle rather than a quadratic witness scan, and measure whether the residual mass continues to decay to zero.

10 A logarithmic frontier oracle

The witness scan of theorem 7.3 runs in $\Theta(Q_{28}^2/M)$ steps, the length of the residue interval (26). The rotation form of theorem 6.2 replaces it, for counting purposes, by a logarithmic computation that moreover reaches heights far beyond the range of any enumeration.

Theorem 10.1 (Floor-sum count). *For every integer $N \geq 1$ with $N < M$,*

$$\#\{1 \leq t < N : \delta^+(t) < t/M\} = S\left(\alpha + \frac{1}{M}, N\right) - S(\alpha, N), \quad S(\beta, N) = \sum_{t=1}^{N-1} \lfloor t\beta \rfloor.$$

Each floor sum is evaluated exactly in $O(\log N)$: if p_j/q_j is a convergent of the real β with $q_j \geq N$, then $\lfloor t\beta \rfloor = \lfloor tp_j/q_j \rfloor$ for all $1 \leq t < N$, and the rational sum $\sum_{t < N} \lfloor tp_j/q_j \rfloor$ is returned by the Euclidean floor-sum recursion.

Proof. For $t \geq 1$ one has $\lceil t\alpha \rceil = \lfloor t\alpha \rfloor + 1$, so $\lfloor t(\alpha + 1/M) \rfloor - \lfloor t\alpha \rfloor = 1$ exactly when the integer $\lfloor t\alpha \rfloor + 1$ lies in $(t\alpha, t\alpha + t/M]$, i.e. when $\lceil t\alpha \rceil - t\alpha = \delta^+(t) < t/M$ (using $t/M < 1$, so the interval holds at most one integer); summation over $t < N$ gives the identity. For the evaluation, $|t\beta - tp_j/q_j| < t/(q_j q_{j+1}) < 1/q_{j+1}$, which is smaller than the distance $\geq 1/q_j$ from tp_j/q_j to $\mathbb{Z}$ whenever $1 \leq t < N \leq q_j$; hence the two floors coincide. The released oracle computes each rational floor sum by the standard subtractive recursion and re-evaluates it at the next convergent as an exactness check. $\square$

Proposition 10.2 (Extended frontier). *The oracle reproduces $\#\mathcal{A}_{28} = 280,139$ and the subcertificate value $\#\mathcal{A}_{26} = 3,867$ exactly, and certifies the admissible counts at heights where the scan (26) is infeasible. The verifier checks that the largest displayed height, Q_{40} , is still below M , so the floor-sum identity of theorem 10.1 applies throughout the table:*

J	Q_J	$\#\mathcal{A}_J$
28	52,449,289,519,716	280,139
29	431,166,034,846,567	18,931,398
30	483,615,324,366,283	23,817,367
32	6,234,549,927,241,963	3,958,255,375
34	267,118,416,222,671,843	7,266,098,126,207
36	4,242,721,909,926,539,673	1,833,085,533,337,202
40	50,247,984,153,525,417,450	257,117,053,950,509,254

The entry at $J = 29$ is reconfirmed independently of the floor-sum oracle by the shipped direct witness scan `tests/independent_a29_scan.py`, which classifies all 37,862,796 residue witnesses with the exact rational enclosures (no floating point), reporting zero undecided cases.

Remark 10.3. Extending the certified frontier broadens the range of periods covered but weakens the conditional height target: the closed form of theorem 7.8 gives $x_0 < Q_{J-1}/(3 \|Q_{J-1}\alpha\|_{\mathbb{Z}} \log 2)$, which grows from 2^{88} at $J = 28$ to 2^{96} at $J = 30$ and 2^{114} at $J = 34$. The oracle is therefore most useful for exact frontier statistics and for the cycle-length floor $\tau(X)$ of theorem 7.6, both available at any height in logarithmic time, independent of the $\Theta(Q^2/M)$ enumeration.

11 An exact affine-ladder certificate for the case $m = 96$

We now apply the prefix theorem to the classical m -cycle decomposition. This section uses the shortcut Collatz map

$$C(n) = \begin{cases} n/2, & n \equiv 0 \pmod{2}, \\ (3n+1)/2, & n \equiv 1 \pmod{2}. \end{cases}$$

A nontrivial periodic orbit is an m -cycle if it has exactly m local minima in one period. Write these minima cyclically as $n_1, \dots, n_m$. After n_i , let k_i be the number of consecutive odd steps and let ℓ_i be the number of following even steps before n_{i+1} . Put

$$K = \sum_{i=1}^m k_i, \quad L = \sum_{i=1}^m \ell_i, \quad \delta = \log_2 3.$$

Thus K is also the odd-state period of the associated accelerated orbit, and the accumulated accelerated valuation over a block prefix is the corresponding sum of the $k_i + \ell_i$.

The proof combines published cycle estimates with an exact finite search. We use the following established inputs.

(I1) Every member of a nontrivial positive cycle is at least

$$X = 2^{71},$$

by Barina's exhaustive verification [1].

(I2) Hercher's Corollary 24 gives

$$m \leq 98 \implies K > 7.76 \cdot 10^{19},$$

and Hercher's Lemma 20 gives $n_{i+1} < n_i^\delta$ for successive local minima [3].

(I3) For $91 \leq m \leq 515,619$, Simons and de Weger give

$$K < 1.4784 m \delta^m. \tag{32}$$

All numerical consequences of these inputs are rechecked by the exact script `code/m96/certify_reduction.py`. The finite-search architecture is closely related to the certified affine-ladder searches introduced by Wang for $m \leq 95$ [5]; the additional gate used here is the Prefix-Hugging frontier of theorem 7.2.

11.1 A suffix-balanced denominator lift

For a local minimum n , define

$$\mathcal{T}(n) = \sum_{j=0}^{k-1} \frac{1}{C^j(n)},$$

where k is the odd-run length beginning at n . The standard cycle product estimate, in Hercher's notation, is

$$0 < \frac{K+L}{K} - \delta < \frac{1}{3K \log 2} \sum_{i=1}^m \mathcal{T}(n_i), \quad \mathcal{T}(n_i) < \frac{3}{n_i}. \tag{33}$$

Lemma 11.1 (Suffix-balanced rotation). *Let $k_1, \dots, k_m$ be positive real numbers with sum K . After a cyclic rotation, one has*

$$\sum_{j=m-t+1}^m k_j \geq \frac{tK}{m} \quad (1 \leq t \leq m). \quad (34)$$

Proof. Set $d_i = k_i - K/m$, so $\sum_i d_i = 0$. Choose a cyclic origin immediately after a maximum of the cyclic partial sums of the d_i . Every prefix sum in the resulting order is nonpositive. Since the total sum is zero, every suffix sum is nonnegative, which is (34). $\square$

Hercher's growth lemma and $2^{k_i} \mid n_i + 1$ imply

$$k_{j+r} < \delta^r \log_2(n_j + 1) \quad (r \geq 1), \quad k_j \leq \log_2(n_j + 1).$$

Consequently, for the rotation in theorem 11.1,

$$\log_2(n_{m-t+1} + 1) \geq \frac{tK}{m} \frac{\delta - 1}{\delta^t - 1}. \quad (35)$$

This gives a convenient exact denominator lift. Let $K_* > 0$, let X_0 be a strict lower bound for all local minima, and choose $0 \leq s \leq m$. Put

$$\bar{\delta} = \frac{317}{200}, \quad \underline{\log 2} = \frac{693147}{10^6},$$

and, for $1 \leq t \leq s$, let f_t be the greatest integer strictly below

$$\frac{tK_*}{m} \frac{\bar{\delta} - 1}{\bar{\delta}^t - 1}.$$

Since $(x - 1)/(x^t - 1)$ decreases for $x > 1$, (33) and (35) give

$$0 < \frac{K + L}{K} - \delta < E(K_*, X_0, s) := \frac{1}{3K_* \underline{\log 2}} \left(\frac{3(m - s)}{X_0} + \sum_{t=1}^s \frac{3}{2^{f_t} - 1} \right) \quad (36)$$

whenever $K \geq K_*$. This estimate involves rational arithmetic only.

For $m = 96$, $K_* = 77,600,000,000,000,000$, $X_0 = X$, and $s = 88$, the exact certificate encloses the interval in (36) between the adjacent fractions

$$\frac{202780263237295321099}{127940101513462006853} \quad \text{and} \quad \frac{123139092617126647266}{77692117359936589403}.$$

Their cross-determinant is one. Every reduced rational strictly between them therefore has denominator at least the denominator of their mediant. Since $(K + L)/K$ lies in this interval, its reduced denominator divides K , and hence

$$K \geq K_0 := 205632218873398596256. \quad (37)$$

The rational logarithm enclosures, determinant, interval containment, and mediant denominator are all part of `certificates/m96_reduction_certificate.json`.

11.2 Finite affine-ladder search

Rotate the hypothetical cycle so that n_1 is its least local minimum. The finite search treats the interval

$$X \leq n_1 \leq 29X. \quad (38)$$

Write

$$n_i = 2^{k_i} a_i - 1, \quad a_i \text{ odd}. \quad (39)$$

The exact block transition is

$$n_{i+1} = \frac{3^{k_i} a_i - 1}{2^{\ell_i}}, \quad \ell_i = v_2(3^{k_i} a_i - 1), \quad k_{i+1} = v_2(n_{i+1} + 1). \quad (40)$$

Since $29X + 1 < 2^{76}$, (39) gives $1 \leq k_1 \leq 75$. Thus the 75 branches $k_1 = 1, \dots, 75$ are a disjoint and exhaustive cover of (38).

Hercher's growth lemma gives

$$k_i < \delta^{i-1} \log_2(n_1 + 1) < \left(\frac{317}{200}\right)^{i-1} 76.$$

The search therefore uses the certified caps

$$(k_1, \dots, k_7) \leq (75, 120, 191, 303, 481, 763, 1210). \quad (41)$$

Combining (37), these caps, and the same geometric growth estimate also yields the stage bounds

$$n_6 \geq 3 \cdot 2^{74}, \quad n_7 \geq 7 \cdot 2^{117}, \quad n_8 \geq \left\lceil \frac{93}{50} 2^{189} \right\rceil. \quad (42)$$

For example, after subtracting the first $i - 1$ caps from K_0 , the remaining $r = 97 - i$ blocks give

$$\log_2(n_i + 1) \geq \left(K_0 - \sum_{j < i} \kappa_j \right) \frac{\delta - 1}{\delta^r - 1},$$

and the exact rational verifier proves that the right side exceeds the logarithm of the corresponding bound in (42).

There is one further necessary prefix condition. At the end of a block prefix, let

$$u = \sum k_i, \quad v = \sum \ell_i.$$

In accelerated notation the prefix has length u and valuation sum $u + v$. If $u + v - u\delta > 0$, theorems 7.2 and 7.3 imply $u \geq 72,057,431,991$. The search has $u \leq 75 + 120 + 191 + 303 + 481 + 763 + 1210 = 3143$, so every realized search prefix must satisfy

$$u + v \leq \lfloor u\delta \rfloor. \quad (43)$$

The C++ kernel evaluates this floor through a certified rational bracket and aborts rather than guessing if the bracket does not decide it.

The search state represents the surviving values of a_1 as one residue class modulo a power of two intersected with an integer interval. For each level it performs two exact partitions. First, the loop over ℓ_i retains a value of a_1 only in the branch equal to the true valuation $v_2(3^{k_i} a_i - 1)$. Second, because the affine coefficient of a_1 is odd, one-bit Hensel lifting partitions that branch by the exact value of $k_{i+1} = v_2(n_{i+1} + 1)$. The interval cuts enforce $n_{i+1} \geq n_1$, (42), and the window floor X ; the congruence cuts enforce (40); and (43) is checked at each completed block. If a live progression contains at most 256 values, the program simulates every value individually with exact GMP arithmetic. The proof of this partition invariant is supplied in docs/m96/FIXED_BRANCH_COMPLETENESS.md.

Theorem 11.2 (Finite-window certificate). *There is no nontrivial 96-cycle whose least local minimum satisfies (38).*

Proof. The task manifest contains exactly one task for each $k_1 \in \{1, \dots, 75\}$. For every task the exact search terminates with $\text{HITS}=\emptyset$. The independent verifier checks the cover, source hash, command line, exit status, raw-log hash, unique result marker, branch range, and zero-hit counter for all 75 tasks. By the partition invariant above, a hypothetical cycle in (38) would survive in exactly one branch and would create a hit. The verified zero-hit manifest therefore excludes the full window. $\square$

11.3 Closing the case $m = 96$

If a 96-cycle remained after theorem 11.2, then its least local minimum, and hence every local minimum, would exceed $29X$. Reapply (36) with $K_* = K_0$, $X_0 = 29X$, and $s = 90$. The resulting interval lies strictly between the adjacent fractions

$$\frac{12261796429850908150604}{7736332199829210068325} \quad \text{and} \quad \frac{325919355854421968365}{205632218873398596256}.$$

Their mediant has denominator

$$Q = 7941964418702608664581, \tag{44}$$

so the same denominator argument gives $K \geq Q$. On the other hand, (32) and $\delta < 317/200$ give, by exact integer comparison,

$$K < 1.4784 \cdot 96 \delta^{96} < 1.4784 \cdot 96 \left(\frac{317}{200}\right)^{96} < Q.$$

This contradiction proves the following theorem.

Theorem 11.3 (Exclusion of 96-cycles). *There is no nontrivial positive Collatz m -cycle with $m = 96$.*

Corollary 11.4. *Together with the certified exclusion of $m \leq 95$ in [5], every nontrivial positive Collatz cycle has at least 97 local minima.*

Remark 11.5. Kevin Knight's exclusion of positive integer high cycles [4] is independent of the argument above. No high-cycle classification is used in the branch cover or in the finite certificate.

12 Exact certificate architecture

The computations in theorems 7.3 and 11.2 are finite theorems with a small trust base. This section records the proof contract for both certificate families.

Lemma 12.1 (Rational logarithm enclosure). *For $x > 1$, set $z = (x - 1)/(x + 1)$ and*

$$L_N(x) = 2 \sum_{j=0}^{N-1} \frac{z^{2j+1}}{2j+1}.$$

Then

$$L_N(x) < \log x < L_N(x) + \frac{2z^{2N+1}}{(2N+1)(1-z^2)}. \tag{45}$$

All endpoints in (45) are rational when x is rational.

Proof. This is the positive series $\log x = 2 \operatorname{artanh} z$. In the omitted tail, $2j + 1 \geq 2N + 1$, so comparison with a geometric series gives the upper bound. $\square$

The release uses $N = 192$ for $x = 2$ and $x = 3$, then rounds outward to 256-bit dyadic endpoints. From these intervals it derives outward rational enclosures for α , ε , and $1/M$. The exact endpoints are recorded in the summary certificate. The certificate and verifier use only the Python standard library, principally arbitrary-precision integers and `fractions.Fraction`.

The one-command verifier performs the following checks.

- (1) It validates an immutable SHA-256 manifest and rejects unlisted files.
- (2) It certifies the first 29 continued-fraction coefficients of α by interval continued fractions and reconstructs every P_j/Q_j through $j = 28$.
- (3) It checks the first-spike range $2 \leq m \leq 102$ by exact integer inequalities and verifies the exact threshold comparison used for $m \geq 103$.
- (4) It parses the A_{28} table defensively, rejecting duplicate times, duplicate witnesses, malformed fields, and noncanonical ordering.
- (5) It scans every $1 \leq d \leq 560,277$, verifies both strict inequalities in (26), and proves that a row is present if and only if the corresponding residue is admissible.
- (6) It checks the decision margin, band totals, extrema, minimum spacing, uniform ρ bound, and the four maximizing times.
- (7) It verifies the height-independent descent-cover certificates: exact residue tiling, prefix-freeness, forced valuations, affine descent data, coverage fractions, and the residual counts used in theorem 8.3.
- (8) It regenerates the frontier, continued-fraction, first-spike, structural-oracle, and descent-cover certificates in a temporary directory and requires byte-for-byte equality. The expensive $m = 96$ branch logs are shipped as raw immutable evidence and are separately reproducible from source.
- (9) It runs adversarial unit tests for hash mutation, extra files, duplicate rows, duplicate witnesses, malformed headers, noncanonical integers, unsafe manifest paths, and unsorted certificates.
- (10) It cross-checks the witness scan against an independent structural oracle: the floor-sum identity of theorem 10.1 recomputes $\#A_{26}$, $\#A_{27}$, and $\#A_{28}$ from certified convergents alone, the cycle-length floor $\tau(2^{71})$ of theorem 7.6 and the minimum gap Q_{16} of theorem 7.4 are reproduced, and the extended counts of theorem 10.2 are regenerated.
- (11) As a non-authoritative regression check, an independent 100-digit `Decimal` implementation reclassifies all 560,277 witnesses without importing the exact interval module.
- (12) It reruns the exact analytic reduction for $m = 96$: logarithm enclosures, continued-fraction brackets, Farey determinants, valuation caps, stage bounds, the finite search window, and the final Simons–de Weger contradiction. The regenerated JSON must equal the shipped analytic certificate byte for byte.
- (13) It verifies the complete branch manifest $k_1 = 1, \dots, 75$ and every raw run record. It checks the C++ source hash, exact command line, successful exit, absence of timeouts, raw-log and metadata hashes, unique summary/result lines, reported branch range, and `HITS=0`; it then recomputes a canonical aggregate digest over all 75 branches.
- (14) It runs mutation tests showing that the branch verifier rejects a missing branch, an altered hit count, a modified log, a source mismatch, an unexpected file, and malformed metadata.

No random test, numerical tolerance, or machine-dependent floating-point mode enters a mathematical decision. The decimal audit is supplementary only; the proof decisions are the exact integer and rational checks. The complete theorem-to-certificate map is included in the release archive.

13 Scope and limitations

The prefix and segment inequalities are unconditional consequences of a hypothetical minimum cycle. The numerical frontier additionally uses the published verification floor (22). Within that scope, the certificate is exhaustive and independently regenerated.

The surplus-frontier and descent conclusions are reductions to explicit finite or structural targets, while theorem 11.3 is a completed cycle exclusion. The remaining limitations are:

- Membership in $\mathcal{A}_{28}$ is necessary, not sufficient. It does not by itself produce a valuation word, satisfy the closing divisibility (3), or establish positivity and primitivity.
- The finite minimum window (28) identifies the exact height range relevant to periods below Q_{28} , and the descent covers remove most residue cylinders inside that range. The residual classes remain genuine unresolved targets.
- The first-spike theorem removes only the branch $2^{m+h} > 3^m$. The remaining branch is constrained by the endpoint quotients but not eliminated here.
- Periods $p \geq Q_{28}$ are outside the finite witness certificate. The floor-sum oracle gives exact frontier statistics at larger convergent heights, but those statistics are still necessary conditions rather than cycle exclusions.
- The cycle-length bound theorem 7.6 is the classical continued-fraction consequence of $\delta^+(p) < p/M$, included because the same oracle evaluates it exactly. It is not presented as a record lower bound.
- The descent-closure statement in Conjecture 9.1 is a conjectural experimental target suggested by the certificates. It is not used as an input to any theorem in this paper.

The package therefore has two roles. For periods below Q_{28} it gives a reproducible reduction to certified surplus times and residual minimum classes. For the separate case $m = 96$, the analytic reduction, exhaustive branch cover, and zero-hit certificate close the argument completely.

Data availability statement

The supplementary release contains the LaTeX source and compiled PDF; the A_{28} , continued-fraction, first-spike, structural-frontier, and descent-cover certificates; the exact $m = 96$ analytic certificate; the 75 raw branch logs and their metadata; the C++ search source; independent Python verifiers; adversarial tests; and an immutable SHA-256 manifest. Running `python3 -B verify_all.py` from the release root checks every shipped certificate without rerunning the expensive branch search. The command in `docs/REPRODUCIBILITY.md` regenerates the branch logs from source.

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